LangChain PyPDFLoader Experiment Document

1. Introduction to the Sun

The Sun is the star at the center of the Solar System. It is a nearly perfect ball of hot plasma, heated to incandescence by nuclear fusion reactions in its core. The Sun radiates this energy mainly as light, ultraviolet, and infrared radiation, providing the primary source of energy for life on Earth.

Its radius is about 695,000 kilometers (432,000 miles), or 109 times that of Earth. Its mass is about 330,000 times that of Earth, accounting for about 99.86% of the total mass of the Solar System. Roughly three-quarters of the Sun's mass consists of hydrogen (~73%); the remainder is mostly helium (~25%), with much smaller quantities of heavier elements, including oxygen, carbon, neon, and iron.

Key Concept for PyPDFLoader:
When loaded via `PyPDFLoader("sample_experiment.pdf")`, this page will be instantiated as `docs[0]` with metadata attribute `{'source': 'sample_experiment.pdf', 'page': 0}`.

2. Basic Physical Characteristics

Property	Value	Earth Comparison
Mean Radius	696,342 km	109 × Earth
Mass	1.9885×10^{30} kg	333,000 × Earth
Surface Gravity	274 m/s^2	28 × Earth
Core Temperature	~15.7 million K	N/A

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3. Energy Transport & Core Fusion

Energy from the core is transported outward through radiation and convection zones. The core extends from the center to about 20–25% of the solar radius. It is the only region in the Sun that produces an appreciable amount of thermal energy through fusion: hydrogen nuclei (protons) are converted into helium nuclei through the proton-proton chain reaction.

```
Proton-Proton Chain Reaction Summary:
4 ¹H → ⁴He + 2 e⁺ + 2 νₑ + 26.73 MeV
```

Above the radiative zone lies the convective zone, where heat moves via fluid motion. Plasma rises toward the surface, cools, and sinks back down, creating granular structures on the solar photosphere.

4. Solar Atmosphere and Space Weather

The solar atmosphere consists of the photosphere, the chromosphere, a transition region, and the corona. The corona extends millions of kilometers into space and gives rise to the solar wind, a stream of charged particles that flows throughout the Solar System.

Solar activity such as solar flares and coronal mass ejections (CMEs) can trigger geomagnetic storms on Earth, impacting satellite communications, power grids, and radio operations.

Experimenting with Content Splitting:
Notice how PyPDFLoader treats each page as a discrete document object. You can use `RecursiveCharacterTextSplitter` to further divide `docs[1].page_content` into smaller vector store embeddings.

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5. Python Implementation Guide

Below is the standard Python code snippet used to load and parse this multi-page PDF document using LangChain.

```
from langchain_community.document_loaders import PyPDFLoader

# Initialize loader with local PDF file path
loader = PyPDFLoader("sample_experiment.pdf")

# Synchronously load all pages into memory
docs = loader.load()

print(f"Total Pages Loaded: {len(docs)}") # Output: 3

# Inspect Page 1
print("Page 1 Metadata:", docs[0].metadata)
print("Page 1 Content Preview:", docs[0].page_content[:150])

# Inspect Page 3
print("Page 3 Metadata:", docs[2].metadata)
print("Page 3 Content Preview:", docs[2].page_content[:150])
```

6. Summary of PyPDFLoader Behavior

- **Output Structure:** Returns `List[Document]` where `len(docs)` equals total PDF pages.
- **Metadata Indexing:** Automatically attaches 0-indexed page numbers (`page: 0`, `page: 1`, `page: 2`).
- **Lazy Loading Support:** Supports `loader.lazy_load()` for streaming multi-page PDFs page-by-page.